

## **L8b:** Option Contracts: Call and Put Payoff and Profit

CHEME 5660 Cornell University

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# Objectives for Today

Today opens Unit 3. Units 1 and 2 modeled assets we hold directly; a **derivative** is instead a contract whose cash flows are a function of someone else's price. We read the contract, compute what it pays at expiration, and keep what the contract promises separate from what the investor earns.

## Today's concepts

- **Read an option contract**: underlying, strike  $K$ , expiration  $T$ , exercise style, deliverable  $q$ , direction  $\theta$ , and the premium: which are contractual, which is quoted.
- **Separate payoff from profit**: the expiration payoff of all four positions, the premium adjustment and the convention it hides, the breakeven prices, and the maximum profit and loss of each.
- **Compose positions**: a strategy as a signed sum of legs, with a corner at each strike where the legs do not cancel and an offset set by the net debit or credit.

# Lectures and Examples

## Lecture:

CHEME-5660-L8b-Lecture-IntroductionToDerivativesContracts-Fall-2026.ipynb

## Example 1:

CHEME-5660-L8b-Example-SingleContractPayoffProfit-Fall-2026.ipynb

## Example 2:

CHEME-5660-L8b-Example-CompositeContractsPL-Fall-2026.ipynb

The first example builds all four single-contract positions from a real options chain, sweeps the full domain  $S_T \geq 0$ , converts per-share quantities to per-contract dollars, and verifies every breakeven and bound numerically; the second combines signed legs into vertical spreads, straddles, and strangles.

# Concept Review: From Assets to Contracts Written on Assets

Units 1 and 2 priced and allocated assets we hold **directly**: Treasury cash flows (L2a, L2b), equity growth rates (L3a, L4b), and portfolios built from them (L5b through L7b).

**Carried forward:** the price of the underlying. Let  $S_t \geq 0$  be that price in USD/share at time  $t$ , so  $S_0$  is today and  $S_T$  is expiration.

- **The payoff is contractual, not statistical:** given  $S_T$ , what the contract pays is a formula written into it. No estimation enters that step. Uncertainty enters only through  $S_T$ .
- **The two sides are not symmetric:** one counterparty holds a right, the other carries an obligation. The premium is the price of that asymmetry, and it is paid whether or not the right is ever used.

# What Is a Derivative?

A **derivative** is a contract whose cash flows depend on an underlying quantity: an equity price, a rate, a currency, a commodity, an index, or a credit event. The families differ in who is obligated:

- **Futures and forwards**: *both* sides must transact the underlying later at a set price. Futures are exchange-traded and standardized; forwards are negotiated over the counter.
- **Swaps**: *both* sides must exchange two streams of cash flows. Traded over the counter.
- **Options**: the buyer holds a *right*, and only the seller carries an obligation. Listed equity options are exchange-traded and centrally cleared; an over-the-counter market also exists.

That asymmetry is the subject of the lecture: it makes an option payoff a **kinked** function instead of a straight line, and it is what the premium buys.

# Reading an Option Contract

Seven quantities must be pinned down before any payoff can be computed:

- the **underlying** asset
- $K > 0$ : strike, USD/share
- $T > 0$ : expiration, years
- **exercise style**: European or American
- $q \in \mathbb{N}$ : deliverable, shares/contract
- $\theta \in \{-1, +1\}$ : direction, +1 long, -1 short
- $C_0, P_0 \geq 0$ : call and put premium, USD/share

Only the **premium** is a market quantity: it is quoted and it moves, and L9a is where it comes from. The rest are contractual or chosen when you trade.

- **Exercise style**: European exercises only at  $T$ , American on or before  $T$ . At expiration the two behave identically, so it does not enter today; it affects the premium (L9a, L9b).
- **The deliverable  $q$** : standard contracts use  $q = 100$ , but the specification controls. A premium quoted at 21.58 USD/share is 2,158 USD of cash for one contract.

# Reading an Option Contract: The Cash Flows

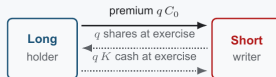
## An option contract: a right and an obligation

The premium is paid once, at the open, whether or not the right is ever used

### Call contract

THE RIGHT TO BUY AT  $K$

- **Long (holder)** pays  $C_0$ /share for the right to buy at  $K$ .
- **Short (writer)** collects  $C_0$ , must deliver at  $K$  if assigned.
- At  $T$ , exercised when  $S_T > K$ : payoff  $(S_T - K)^+$ .

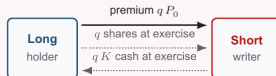


solid: always. dotted: only if  $S_T > K$ .

### Put contract

THE RIGHT TO SELL AT  $K$

- **Long (holder)** pays  $P_0$ /share for the right to sell at  $K$ .
- **Short (writer)** collects  $P_0$ , must buy at  $K$  if assigned.
- At  $T$ , exercised when  $S_T < K$ : payoff  $(K - S_T)^+$ .



solid: always. dotted: only if  $S_T < K$ .

European style exercises only at  $T$ ; American style on or before  $T$ . Commonly  $q = 100$  shares/contract.

One flow is unconditional, the premium at the open; two are conditional on exercise. **Long** positions risk only the premium, **short** ones collect it and carry the obligation.



# Expiration Payoff Is Contractual

For any real  $x$  write  $x^+ \equiv \max(x, 0)$ . With  $S_T \geq 0$  the underlying price at expiration and  $K > 0$  the strike, both in USD/share, the payoff **per share** to the long holder is given by:

$$h_c(S_T; K) = (S_T - K)^+, \quad h_p(S_T; K) = (K - S_T)^+$$

The call holder exercises only when  $S_T > K$ , the put holder only when  $S_T < K$ , and otherwise abandons the right. Because the holder may always decline, **both payoffs are nonnegative**: the signature of a right.

- **The short side mirrors it**: with direction  $\theta \in \{-1, +1\}$ , the signed payoff is  $\theta h_c$  or  $\theta h_p$ .
- **Moneyness** classifies exercise value, not profit: a call is in the money when  $S_T > K$ , out of the money when  $S_T < K$ , and the put reverses these. It says nothing about whether the trade made money.

# Profit Requires a Cash-Flow Convention

Payoff belongs to the contract. **Profit belongs to the investor**, and it needs a rule for the premium, which moved at a different time. Under the **undiscounted convention**, with  $C_0, P_0 \geq 0$  the premiums in USD/share and  $\theta$  the direction, profit per share is given by:

$$\Pi_T^{c,\theta}(S_T) = \theta \left( \underbrace{h_c(S_T; K)}_{\text{payoff at } T} - \underbrace{C_0}_{\text{premium at 0}} \right), \quad \Pi_T^{p,\theta}(S_T) = \theta \left( h_p(S_T; K) - P_0 \right)$$

For a long position the premium is an outflow, for a short an inflow: one  $\theta$  multiplies both terms.

- **Say what it hides**: this mixes time-0 and time- $T$  dollars, which every earlier lecture avoided. A time- $T$  P&L would grow the premium forward, a time-0 value would discount the payoff (L1b).
- **Why use it**: it gets the shape and the breakeven right with the fewest moving parts. A convention, not an accounting identity.

## Profit: Where a Position Breaks Even

The **breakeven**  $S_{\text{BE}}$ , in USD/share, is the terminal price at which profit is exactly zero. Setting  $\Pi_T = 0$  on the exercised branch gives  $S_T - K - C_0 = 0$  for the long call and  $K - S_T - P_0 = 0$  for the long put, hence:

$$S_{\text{BE}}^c = K + C_0, \quad S_{\text{BE}}^p = K - P_0$$

- **The breakeven is not the strike**: the premium shifts the curve down (long) or up (short) without moving the kink at  $K$ . On the AMD chain a  $K = 230$  call at  $C_0 = 21.58$  breaks even at 251.58: finishing in the money is not the same as making money.
- **Both directions share it**: short profit is the exact negative of long profit, so both sides agree on the breakeven price.

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**Example 1:** CHEME-5660-L8b-Example-SingleContractPayoffProfit-Fall-2026.ipynb

# Loss Bounds and Direction Matter

All per share; multiply by  $q$  and by the number of contracts for account dollars.

| Position   | Maximum profit                        | Maximum loss                          | Breakeven |
|------------|---------------------------------------|---------------------------------------|-----------|
| Long call  | unbounded as $S_T \rightarrow \infty$ | $C_0$                                 | $K + C_0$ |
| Short call | $C_0$                                 | unbounded as $S_T \rightarrow \infty$ | $K + C_0$ |
| Long put   | $K - P_0$ at $S_T = 0$                | $P_0$                                 | $K - P_0$ |
| Short put  | $P_0$                                 | $K - P_0$ at $S_T = 0$                | $K - P_0$ |

The difference between the call rows and the put rows is the **domain**:  $S_T \geq 0$ , so  $h_p \leq K$  is capped by the strike, while  $h_c$  has no cap because  $S_T$  has none.

- **Only one loss is genuinely unlimited**: the uncovered short call. The long call's profit is unbounded too, but no loss is.
- **A short put's loss is finite**:  $K - P_0$  at  $S_T = 0$ . On the  $K = 220$  contract collecting 16.33, that is 203.67 USD/share, or 20,367 USD per contract. **Large is not unlimited**.

## Composite Positions Add Leg by Leg

Let position  $\mathcal{C}$  contain legs  $i = 1, \dots, d$  on one underlying and one expiration. Leg  $i$  has strike  $K_i > 0$ , direction  $\theta_i \in \{-1, +1\}$ , copies  $n_i \in \mathbb{N}$ , premium  $P_i \geq 0$  in USD/share, and long per-share payoff  $h_i(S_T)$ . The **per-share** totals are given by:

$$H_{\mathcal{C}}(S_T) = \sum_{i=1}^d \underbrace{\theta_i n_i}_{\text{signed copies of leg } i} h_i(S_T), \quad \Pi_{\mathcal{C}}(S_T) = \sum_{i=1}^d \theta_i n_i [h_i(S_T) - P_i]$$

Both are in USD/share, which is what the example notebooks plot. For account dollars, multiply leg  $i$  by its deliverable  $q_i \in \mathbb{N}$  shares/contract.

- **The kinks are the strikes:** each  $h_i$  bends only at  $K_i$ , so the profit is piecewise linear, with a corner at strike  $K$  exactly when the aggregate jump  $\sum_{i:K_i=K} \theta_i n_i$  is nonzero. Opposing legs at one strike cancel.
- **The offset is the net premium:**  $\sum_i \theta_i n_i P_i$ , the net debit or credit, shifts the diagram without changing its shape.

Legs must share an underlying and an expiration for one diagram in  $S_T$ ; deliverables may differ, so each  $q_i$  is applied to its own leg.

# Composite Positions: Three Structures, One Sum

Three structures,  $D$  the total premium per share:

- **Vertical spread**: two legs of one type, different strikes, opposite directions. Both extremes finite.
- **Straddle**: a call and a put at one strike  $K$ , same direction. Profit  $\theta(|S_T - K| - D)$ , breakevens  $K + D$  and, when  $D \leq K$ ,  $K - D$ .
- **Strangle**: a call and a put at strikes  $K_1 < K_2$ , same direction. A flat plateau replaces the straddle's single kink: a trough when long, a peak when short.

**State the bounds correctly.** A long straddle is routinely said to have “unlimited profit in either direction.” **That is wrong on the downside.** Since  $S_T \geq 0$  the put leg cannot pay more than its strike, so the lower branch terminates: the AMD  $K = 230$  straddle with  $D = 42.88$  caps its downside at 187.12 USD/share, while the upside has no bound.

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**Example 2:** CHEME-5660-L8b-Example-CompositeContractsPL-Fall-2026.ipynb

# Optional Advanced Material

**Advanced:** [advanced/static\\_replication/](#)  
[CHEME-5660-L8b-Advanced-Static-Replication-Fall-2026.ipynb](#)

Optional, not a prerequisite for L9a. The composite rule says adding legs **produces** payoff shapes; the notebook proves the converse. Every continuous piecewise-linear payoff with kinks  $0 < K_1 < \dots < K_d$  is  $a + b S_T + \sum_i w_i (S_T - K_i)^+$  in **exactly one** way, with  $a$  the payoff at zero,  $b$  the initial slope, and  $w_i$  the jump in slope at  $K_i$ . It proves existence by a telescoping sum of slopes and uniqueness by sweeping out from the leftmost interval, derives the put payoff identity, tabulates every structure above as one triple  $(a, b, w)$ , and marks the boundary of what a single expiration can reach.

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**Notebook:** [lectures/week-8/L8b/advanced](#), also linked from the lecture notebook.

# Summary

Today we translated contract terms into expiration cash flows and kept the contractual payoff separate from the investor's premium-dependent profit.

## Key takeaways

- **Payoff belongs to the contract:** a long call pays  $(S_T - K)^+$  and a long put pays  $(K - S_T)^+$  per share, and short positions reverse those flows exactly. Only  $S_T$  is uncertain.
- **Profit needs a convention:** premium, financing time, deliverable  $q$ , fees, and direction  $\theta$  all enter, so strike, moneyness, and breakeven are three different things.
- **Loss bounds are asymmetric, and only one side is unlimited:** the uncovered short call has no finite bound because  $S_T$  has none; a standard short put's worst case is finite at  $K - P_0$ , reached at  $S_T = 0$ .

Next, L9a values European claims with Black–Scholes–Merton and American claims with a Cox–Ross–Rubinstein lattice: where the premium comes from.



Today: what the contract promises, what the investor earns,  
and why only one of the four positions can lose without  
bound.